Board C – Zynq-7020 processing board / FMC carrier

Power: 12 V in, sequenced rails, DDR termination

File: power.kicad_sch

Zynq power pins, configuration, JTAG, XADC

File: zynq_power.kicad_sch

Zynq PS: clock, reset, boot straps, QSPI, SD, UART, I2C

File: zynq_ps.kicad_sch

DDR3L 512 MB x16 (PS DDR, bank 502)

File: ddr3.kicad_sch

Gigabit Ethernet: RTL8211F (RGMII 3.3 V) + HR911130C

File: ethernet.kicad_sch

GNSS: u-blox MAX-M10S, PPS to PL for time sync

File: gnss.kicad_sch

FMC LPC carrier: digitizer LVDS on banks 34/35 (2.5 V)

File: fmc.kicad_sch

Zynq XC7Z020-1CLG400C (all units)

File: zynq.kicad_sch

PWR_FLAG

+12V

PWR_FLAG

+1V35

PWR_FLAG

+3V3

PWR_FLAG

GND

Sheet: /
File: board_c.kicad_sch

Title: Board C – Zynq-7020 processing board / FMC carrier

Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.3		Id: 1/9

Power: 12 V in, sequenced rails, DDR termination

Sequence: 1.0 V (VCCINT/VCCPINT) -> 1.8 V (VCCAUX/VCCPAUX) -> 1.35 V, 3.3 V, 2.5 V -> PS_POR_B.

VIN_RAW
GND
F1 3A
D1 SMBJ15A
C1 22u
C2 22u
TPS62933DRGTR
L1 2.2u
R1 25k
R2 100k
C3 3.3n
C4 10u
C5 10u
C6 100n
C7 22u
C8 22u
C9 100n
R3 100k
R4 47k
L2 2.2u
R5 125k
R6 100k
C10 3.3n
C11 10u
C12 10u
C13 100n
C14 22u
C15 22u
C16 100n
R7 100k
L3 2.2u
R8 68.8k
R9 100k
C17 3.3n
C18 10u
C19 10u
C20 100n
C21 22u
C22 22u
C23 100n
R10 100k
R11 47k
L4 2.2u
R12 312.5k
R13 100k
C24 3.3n
C25 10u
C26 10u
C27 100n
C28 22u
C29 22u
C30 100n
R14 100k
L5 2.2u
R15 212.5k
R16 100k
C31 3.3n
C32 10u
C33 10u
C34 100n
C35 22u
C36 22u
C37 100n
R17 100k
TPS51200DRCR
R18 10k
R19 10k
C38 100n
C39 10u
C40 10u
C41 10u
C42 100n
C43 1u
F2 1A
C44 10u
R20 10k
R21 1k
C45 10u
SW1 PGR
H1 M3
H2 M3
H3 M3
H4 M3
TP1 +1V0
TP2 +1V8
TP3 +1V35
TP4 +3V3
TP5 VADJ
TP6 DDR_VTT

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Passive radar receiver

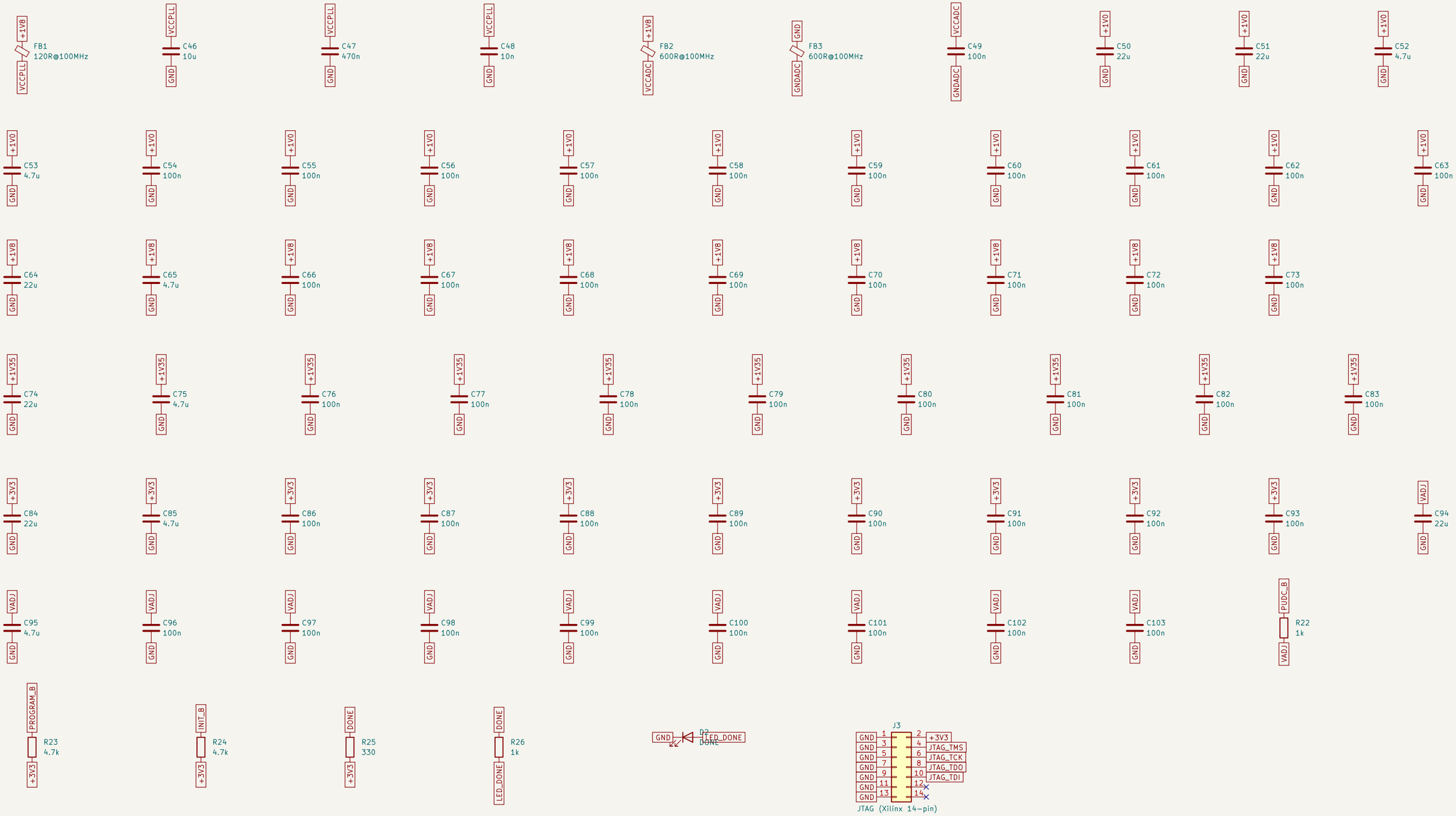
Sheet: /Power: 12 V in, sequenced rails, DDR termination/
File: power.kicad_sch

Title: Power: 12 V in, sequenced rails, DDR termination

Size: User	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 2/9

Zynq power pins, configuration, JTAG, XADC

VCCBATT_0 tied to GND (no battery-backed key). XADC unused: VREFP/VREFN/VP/VN -> GNDADC (internal ref).



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Passive radar receiver

Sheet: /Zynq power pins, configuration, JTAG, XADC/
File: zynq_power.kicad_sch

Title: Zynq power pins, configuration, JTAG, XADC

Size: User	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 3/9

Zynq PS: clock, reset, boot straps, QSPI, SD, UART, I2C

Boot mode MIO[5:3]: QSPI = 1,0,0 (jumper open), SD = 1,1,0 (jumper closed). Console UART1 115200 8N1.

Y1 33.333MHz
OE/NC, GND, VDD, VOUT
+3V3, PS_CLK_OSC
R27 22
C104 100n
PS_SRST_B
R28 4.7k
SW2 SRST
R29 1k
MIO_VREF
R30 1k
QSPI_IO0
R31 20k
R32 20k
QSPI_CLK
R33 20k
STRAP_MIO7
R34 20k
STRAP_MIO8
R35 20k
QSPI_IO2
R36 20k
QSPI_IO3
R37 20k
R38 4.7k
BOOT_SD_J
J4 BOOT: open=QSPI, closed=SD
U9 W25Q256JVEIQ
QSPI_CS, QSPI_IO1, QSPI_IO2, QSPI_IO3, QSPI_CLK, QSPI_IO0
R39 10k
C105 100n
SD_CLK_MIO
R40 33
SD_CLK
J5 microSD
SD_D0, SD_D1, SD_D2, SD_D3, SD_CMD, SD_CLK, SD_CD
R41 10k
SD_CMD
R42 10k
SD_D0
R43 10k
SD_D1
R44 10k
SD_D2
R45 10k
SD_D3
R46 10k
SD_CD
C106 10u
C107 100n
I2C0_SCL
R47 4.7k
I2C0_SDA
R48 4.7k
LED_PS
R49 1k
LED_PS_A
D7 LED_PS_A
R50 10k
USER_BTN_PS
SW3 USER
J6 USB_VBUS, USB_CC1, USB_CC2, USB_DN, USB_DP
R51 5.1k
USB_CC1
R52 5.1k
USB_CC2
U10 CP2102N
USB_VBUS_SENSE, CP_RSTB, UART_RX, UART_TX
R53 22.1k
USB_VBUS
R54 47.5k
USB_VBUS_SENSE
R55 4.7k
CP_RSTB
C108 1u
C109 100n
C110 1u

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Passive radar receiver

Sheet: /Zynq PS: clock, reset, boot straps, QSPI, SD, UART, I2C/
File: zynq_ps.kicad_sch

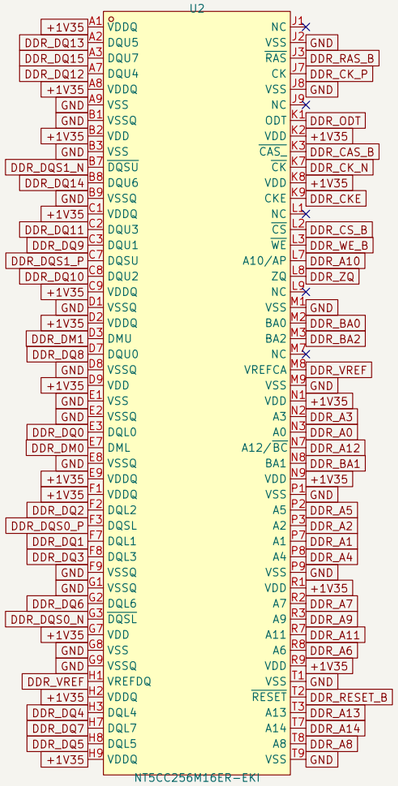
Title: Zynq PS: clock, reset, boot straps, QSPI, SD, UART, I2C

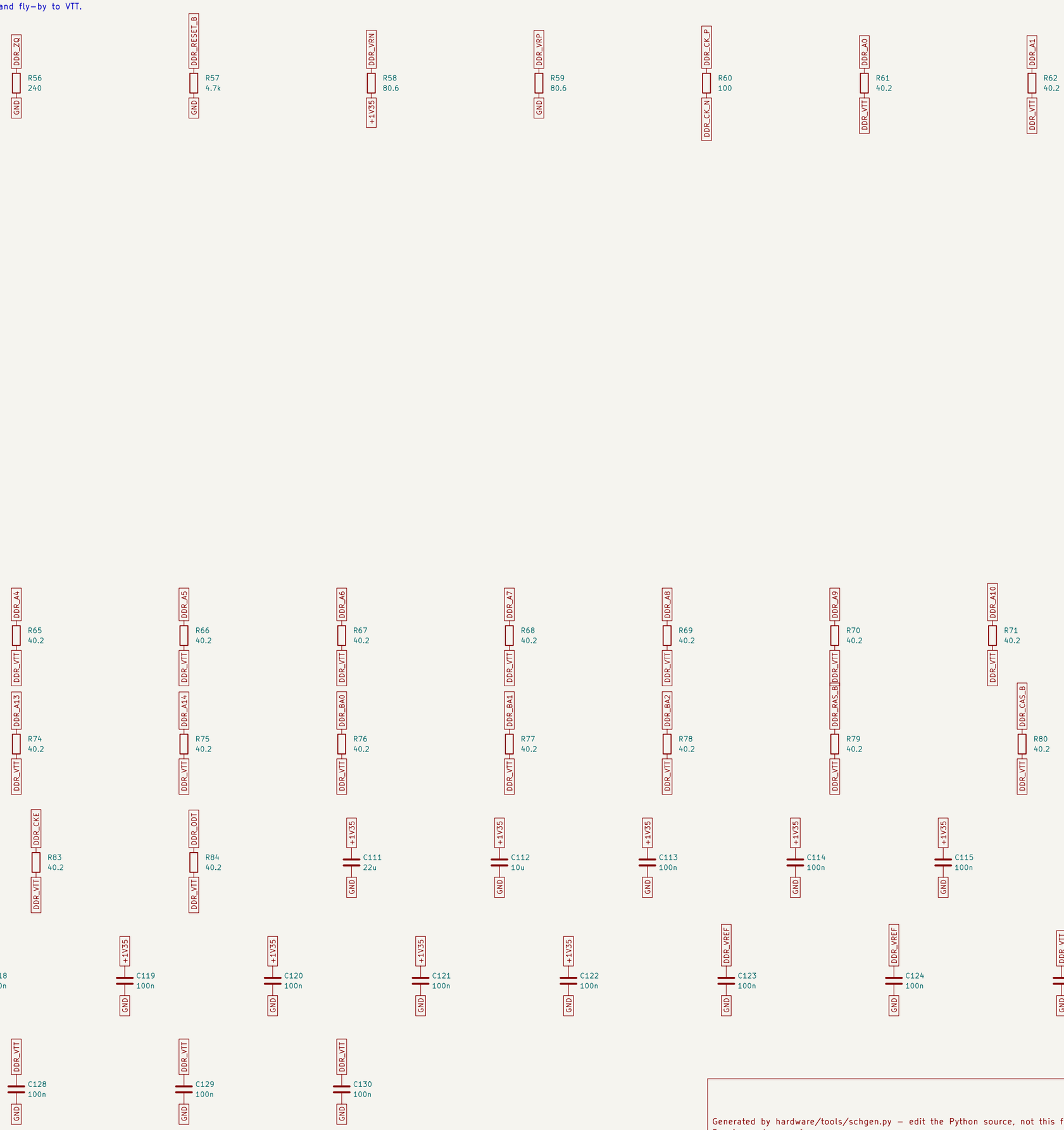
Size: User	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 4/9

DDR3L 512 MB x16 (PS DDR, bank 502)

16-bit configuration: PS_DDR DQ16-31, DM2/3, DQS2/3 left unconnected (verify with Xilinx UG933).

Route DQ byte lanes length-matched to their DQS (+/- 0.5 mm); address/command fly-by to VTT.





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Passive radar receiver

Sheet: /DDR3L 512 MB x16 (PS DDR, bank 502)/

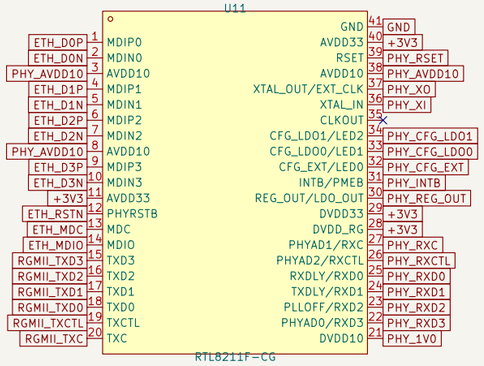
File: ddr3.kicad_sch

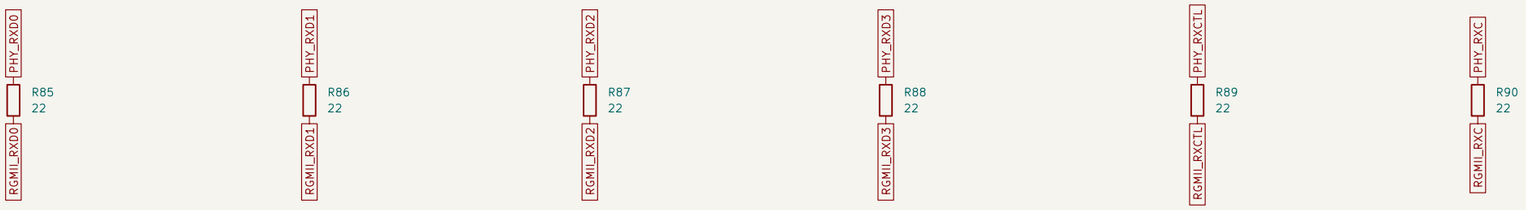
Title: DDR3L 512 MB x16 (PS DDR, bank 502)

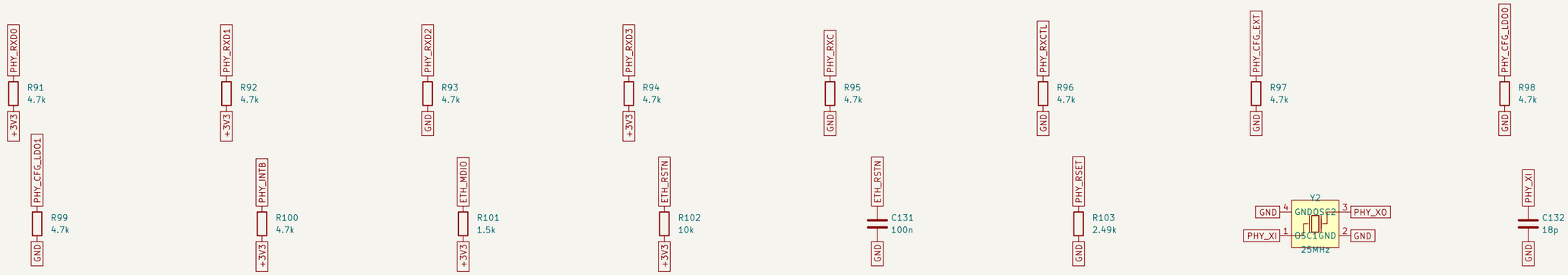
Size: User	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 5/9

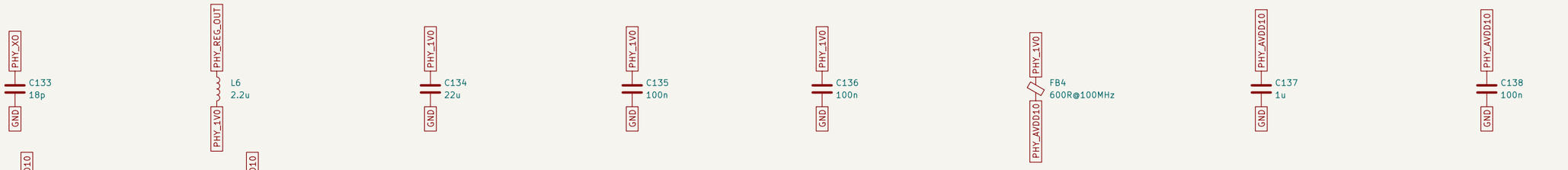
Gigabit Ethernet: RTL8211F (RGMII 3.3 V) + HR911130C

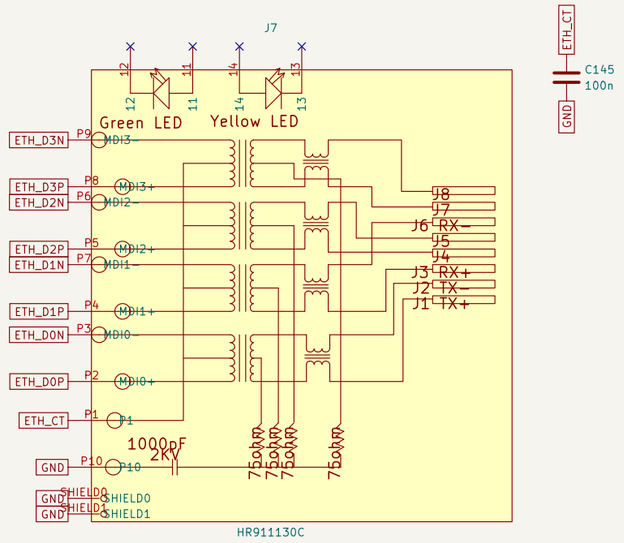
Magjack P1 = centre taps (AC-decoupled; RTL8211F is a voltage-mode driver), P10 = GND. LEDs unused. VERIFY.











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Passive radar receiver

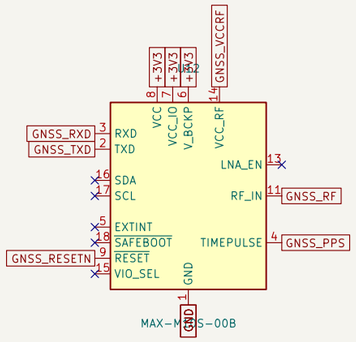
Sheet: /Gigabit Ethernet: RTL8211F (RGMII 3.3 V) + HR911130C/
File: ethernet.kicad_sch

Title: Gigabit Ethernet: RTL8211F (RGMII 3.3 V) + HR911130C

Size: User	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 6/9

GNSS: u-blox MAX-M10S, PPS to PL for time sync

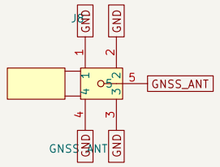
Active antenna powered from VCC_RF via 10 ohm + 47 nH. PPS -> bank 13 SRCC pin for timestamping.



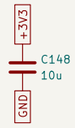
























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Passive radar receiver

Sheet: /GNSS: u-blox MAX-M10S, PPS to PL for time sync/
File: gnss.kicad_sch

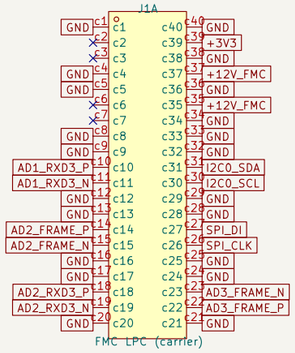
Title: GNSS: u-blox MAX-M10S, PPS to PL for time sync

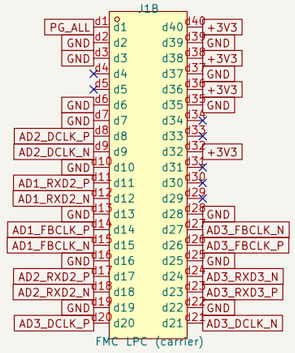
Size: User	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 7/9

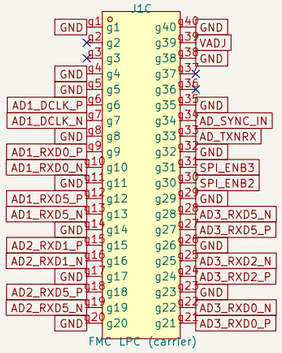
FMC LPC carrier: digitizer LVDS on banks 34/35 (2.5 V)

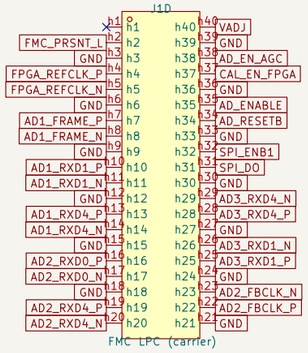
LVDS pairs: 100 ohm differential, all 9 pairs of each AD9361 matched to +/− 1 mm.

Clock-capable pins: AD1_DCLK->B35.L12, AD2_DCLK->B35.L13, AD3_DCLK->B34.L12, FPGA_REFCLK->B35.L11 (SRCC).

















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Passive radar receiver

Sheet: /FMC LPC carrier: digitizer LVDS on banks 34/35 (2.5 V)/

File: fmc.kicad_sch

Title: FMC LPC carrier: digitizer LVDS on banks 34/35 (2.5 V)

Size: User	Date:	**Rev:**
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Zynq XC7Z020-1CLG400C (all units)



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Passive radar receiver

Sheet: /Zynq XC7Z020-1CLG400C (all units)/
File: zynq.kicad_sch

Title: Zynq XC7Z020-1CLG400C (all units)

Size: User	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 9/9